

AMERICAN TERAWATT

HVDC Design

American Terawatt | July 30, 2026

RAM

Reliability, Availability and Maintenance

Performance Reporting

Availability definitions & formulas — CIGRE TB 956

Scope of reporting

CIGRE TB 956 (2025) — supersedes TB 590

A standardized protocol for reporting the annual operational performance (reliability, availability, maintenance) of HVDC systems.

- Reporting period is the calendar year (Jan–Dec).
- Covers point-to-point links, back-to-back, multi-terminal systems and DC grids (LCC and VSC).
- Only events causing a loss of transfer capability are counted; AC-system outages are recorded but excluded from HVDC reliability figures.

Outage classifications

Outage

A state in which the HVDC system/station is unavailable at its maximum continuous capacity due to an event related to the converter station or DC line.

- **Scheduled outage** — planned, or deferrable to a suitable time (e.g. preventive maintenance). Reported as “PM”.
- **Forced outage** — equipment unavailable but not in a scheduled-outage state.
 - *Trips* — sudden interruption by automatic protection or manual emergency shutdown.
 - *Other forced outages* — unexpected problems forcing a capacity reduction without a trip.
- **Forced Outage Rate (FOR)** - Ratio of the number of all complete or partial outages of one converter (or one pole in a bipolar configuration, respectively) to the whole duration of operation.

Rated, outage capacity & derating factor

- **Rated capacity** P_m — maximum capacity (MW) for continuous operation under normal conditions, excluding redundant equipment. If seasonal, the highest value is used.
- **Outage capacity** P_o — the capacity reduction (MW) the outage would have caused if the system were operating at P_m .
- **Outage derating factor (ODF)** — ratio of outage capacity to rated capacity:

$$\text{ODF} = \frac{P_o}{P_m}$$

A full outage gives $\text{ODF} = 1$; a partial loss gives $0 < \text{ODF} < 1$.

Outage duration

- **Period Hours (PH)**

Calendar hours in the period: 8760 h/yr (8784 in a leap year)

- **Actual outage duration (AOD)** — time elapsed (decimal h) between the start and end of an outage, from the first switching action to the return of the equipment to operational readiness.

- **Equivalent outage duration (EOD)** — AOD weighted by the derating factor, to account for partial loss of capacity:

$$\text{EOD} = \text{AOD} \times \text{ODF}$$

Classified by type as equivalent forced (EFOD)
or equivalent scheduled (ESOD) outage duration.

Time categories

Actual Outage Hours

- **Actual Outage Hours (AOH)**

The sum of actual outage durations within the reporting period

$$AOH = \sum AOD$$

- **Actual forced outage hours (AFOH)**

The sum of equivalent forced outage durations within the reporting period

$$AFOH = \sum AFOD$$

- **Actual scheduled outage hours (ASOH)**

The sum of equivalent scheduled outage durations within the reporting period

$$ASOH = \sum ASOD$$

Time categories

Equivalent Outage Hours

- **Equivalent Outage Hours (EOH)**

The sum of equivalent outage durations within the reporting period

$$EOH = \sum EOD$$

- **Equivalent Forced Outage Hours (EFOH)**

The sum of equivalent forced outage durations within the reporting period

$$EFOH = \sum EFOD$$

- **Equivalent scheduled outage hours (ESOH)**

The sum of equivalent scheduled outage durations within the reporting period

$$ESOH = \sum ESOD$$

Energy unavailability & availability

- **Energy unavailability (EU)**

Percentage of energy that could not be transmitted due to outages:

$$EU (\%) = \frac{EOH}{PH} \times 100$$

- **Forced Energy Unavailability (FEU)**

$$FEU = \frac{EFOH}{PH} \times 100$$

- **Scheduled Energy Unavailability (SEU)**

$$SEU = \frac{ESOH}{PH} \times 100$$

- **Energy availability (EA)**

Complement of unavailability:

$$EA (\%) = 100 - EU$$

Energy utilization

- **Energy utilization** U — measure of the energy actually transmitted over the system, relative to its maximum possible:

$$U (\%) = \frac{\text{Total Energy Transmitted}}{P_m \times \text{PH}} \times 100$$

- Total energy transmitted = energy exported + energy imported (MWh), referred to the point at which P_m is defined.

HVDC projects

Guarantees and Liquidated damages

Typical values for a HVDC project

Metric	Description	Typical value	Guaranteed
FEU	Forced Energy Unavailability	0.2 - 1.0 %	0.5 %
SEU	Scheduled Energy Unavailability	1 - 3 %	1.0 %
FOR	Forced Outage Rate (pole)	$\approx 4 - 8 / \text{yr}$	2 - 3 / yr
	Forced Outage Rate (bipole)	$\lesssim 0.5 / \text{yr}$	0.1 / yr

- Indicative ranges from CIGRE biennial HVDC reliability surveys.
- Actual figures vary with technology (LCC vs. VSC), scheme age, and transmission medium — long overhead lines raise forced-outage counts, cable schemes lower them.
- For bipole outages values of 0.05 / y have been guaranteed, but the values do not account for the real performance and it is assumed that liquidated damages are included in commercial risk.

Liquidated damages — FOR & SEU

Metric	Penalty rate
FOR	0.1 - 1.2 Mio. \$
FEU	75 - 600 tsd. \$

- Penalties (liquidated damages) apply when the measured value exceeds the contractually guaranteed threshold.
- Typically assessed per unit of excess
 - FOR: additional forced outage
 - FEU: additional 0.1 % of unavailability
- The cap liquidated damages is in the range of 3-5 % of the contract value

Data Center Availability

Calculation methodology & topology comparison

Abbreviations

CRAH	Computer Room Air Handler
FMEA	Failure Mode and Effects Analysis
MTBF	Mean Time Between Failures
MTTR	Mean Time To Repair
MV / LV	Medium / Low Voltage
PDU	Power Distribution Unit
RBD	Reliability Block Diagram
TIA	Telecommunications Industry Association (TIA-942)
UPS	Uninterruptible Power Supply

What “availability” means

Availability is the fraction of time a facility (or system) can deliver IT load without interruption, measured over a defined period (typically a rolling 12 months).

Core relationship

$$A = \frac{\text{Uptime}}{\text{Uptime} + \text{Downtime}} = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$$

where

MTBF is mean time between failures

MTTR is mean time to repair/recover.

Availability is reported as a percentage or in “nines.”

The language of nines

Availability	Nines	Allowable downtime / yr
99 %	two	$\approx$ 3.65 days
99.9 %	three	$\approx$ 8.77 hours
99.99 %	four	$\approx$ 52.6 minutes
99.999 %	five	$\approx$ 5.26 minutes
99.9999 %	six	$\approx$ 31.5 seconds

Every additional nine cuts allowable downtime by a factor of ten.

Combining components

A facility is modeled as a reliability block diagram (RBD) of components in *series* and *parallel*.

Series — all required

$$A = \prod_i A_i$$

Any element fails $\Rightarrow$ outage.
Always *below* the weakest link.

Parallel — redundant

$$A = 1 - \prod_i (1 - A_i)$$

Survives if ≥ 1 path works.
Redundancy raises A sharply.

Modeling assumptions

Frameworks

- Uptime Institute Tier standard (I–IV)
- TIA-942 infrastructure standard
- RBD, FMEA, Monte Carlo for complex topologies

Scope

- Facility *infrastructure* availability
- Facility = Power path $\times$ Cooling path

Reading the topology diagrams

Redundancy notation

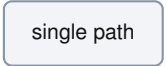
- N — just enough capacity, no spare
- $N+1$ — one spare unit
- $2N$ — fully duplicated

Border legend

- ■ **Double border** = $N+1$ redundant stage
- ■ **Single border** = single (concurrently maintainable) path

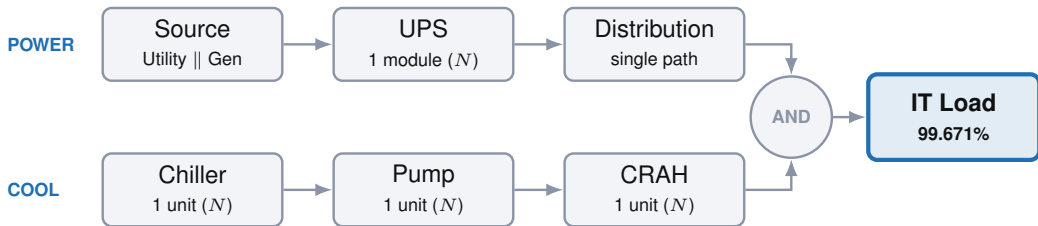
A light blue rounded rectangle with a double blue border.

$N+1$ stage

A light blue rounded rectangle with a single blue border.

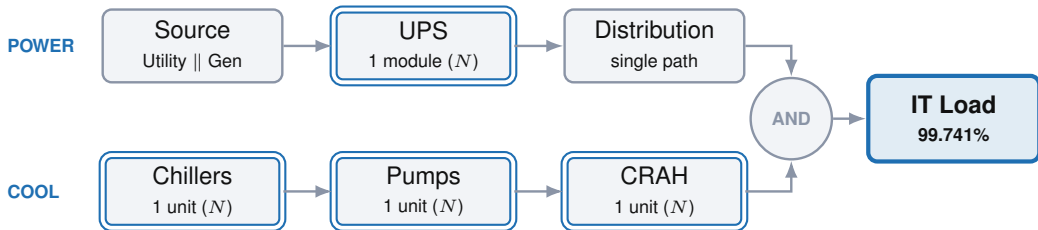
single path

Tier I topology



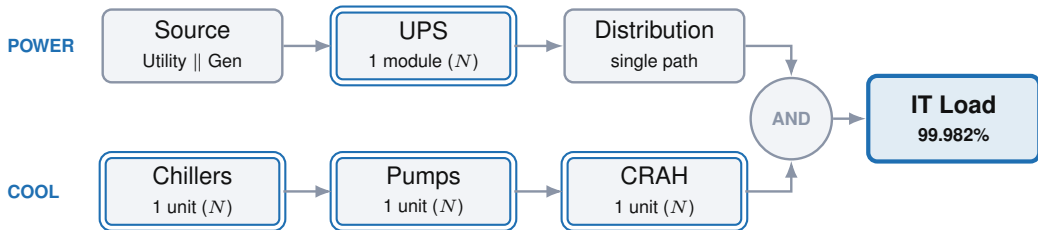
- No redundancy anywhere
- Single distribution path
- *Not* concurrently maintainable - planned maintenance requires a shutdown

Tier II topology



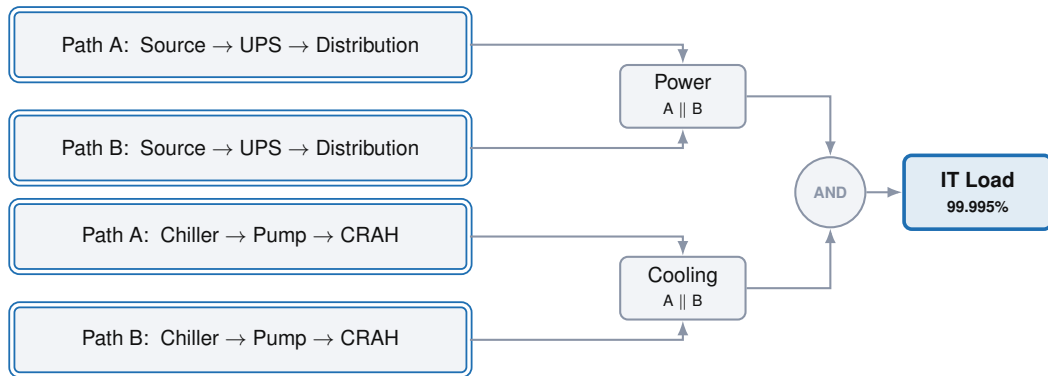
- Redundant capacity components ($N+1$) but a single, non-redundant distribution path
- *Not* concurrently maintainable
- Same component RBD as Tier III, the difference is planned-maintenance downtime

Tier III topology



- Redundant capacity components ($N+1$) but a single, non-redundant distribution path
- Concurrently maintainable
- Same component RBD as Tier III, the difference is planned-maintenance downtime

Tier IV topology



- Every stage duplicated end-to-end ($2N$)
- No single non-redundant link
- Unavailable only if *both* A and B fail

Tier Comparison

	I (N)	II ($N+1$)	III ($N+1$)	IV ($2N$)
Redundancy	none	unit	unit	full path
Distribution	single	single	single	dual A/B
Conc. maint.	no	no	yes	yes
Fault tolerant	no	no	no	yes
Availability	99.671 %	99.741 %	99.982 %	99.995 %
Downtime/yr	≈ 28.8 h	≈ 22 h	≈ 1.6 h	≈ 26.3 min